

Reliable Network Traffic Intelligence – ZOU Xiexin

Project area: traffic data quality, traffic prediction, and pattern-change analysis

1. Research question

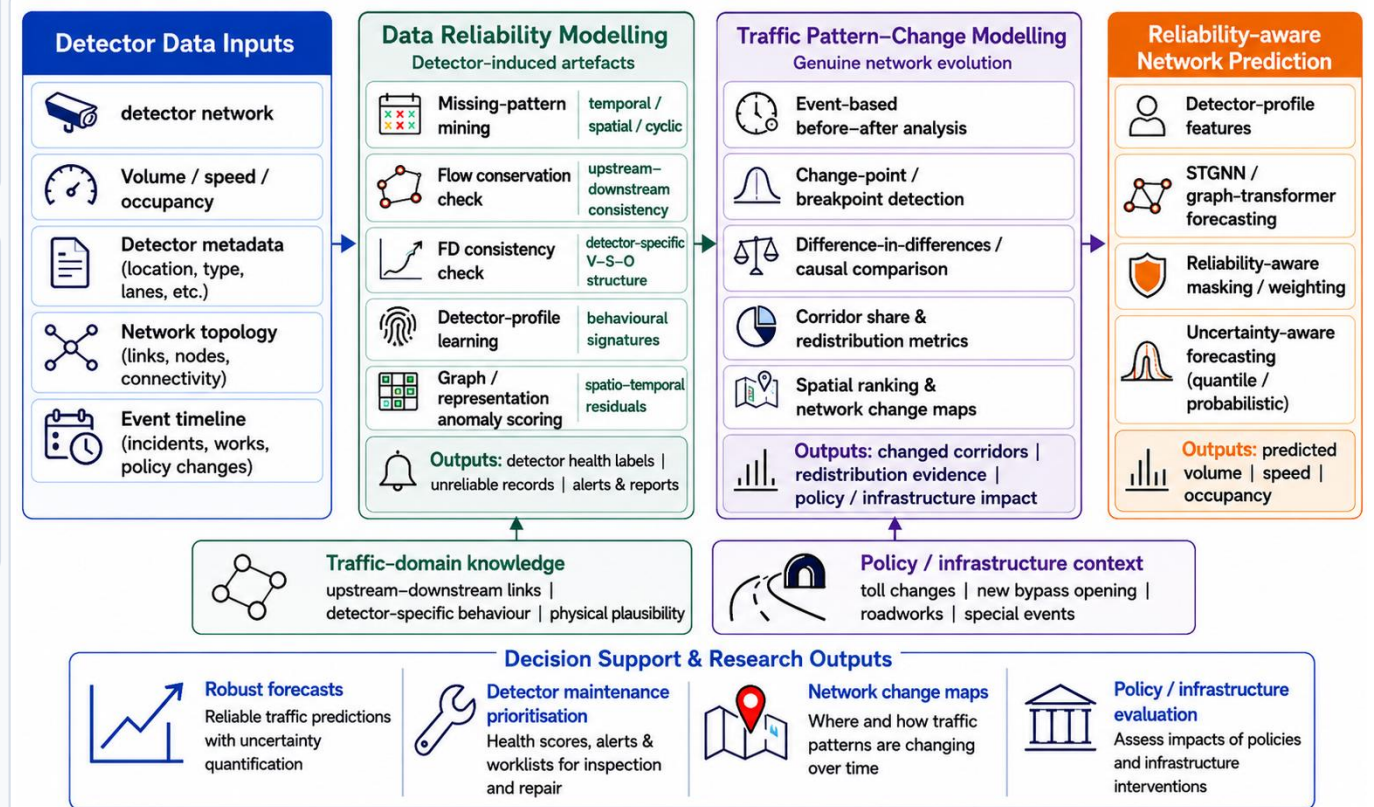
How to distinguish anomalies in detector data from genuine traffic evolution, and use this distinction to improve network-level traffic prediction and policy/infrastructure evaluation?

2. Why it matters

- Reliable traffic prediction depends on data quality, and unreliable inputs can significantly bias operational decisions.
- Long-term traffic changes reveal how the network responds to policies, infrastructure changes, and demand shifts, supporting more reliable evaluation of transport interventions.

3. Contribution

- Domain-informed diagnostic framework for traffic data quality assessment using missing patterns, flow conservation, and FD consistency.
- Reliability-aware network prediction framework integrating detector data quality and learned representations for robust forecasting.
- Pattern-change analysis of long-term traffic redistribution for policy and infrastructure evaluation.



Next step: Conduct network-wide long-term pattern-change analysis to quantify traffic redistribution under policy and infrastructure interventions. Extend reliability-aware prediction with uncertainty modelling to improve forecasting robustness.

Uncertainty-Aware Prediction in Traffic-Related Data – Beiyu Song

Project area: Traffic prediction & uncertainty quantification | Stage: modelling / validation / writing

1. Research aim

This research develops reliable prediction intervals for traffic-related data – OD demand and bus arrival time – so decisions know not only what is likely to happen, but also how uncertain it is.

2. Why it matters

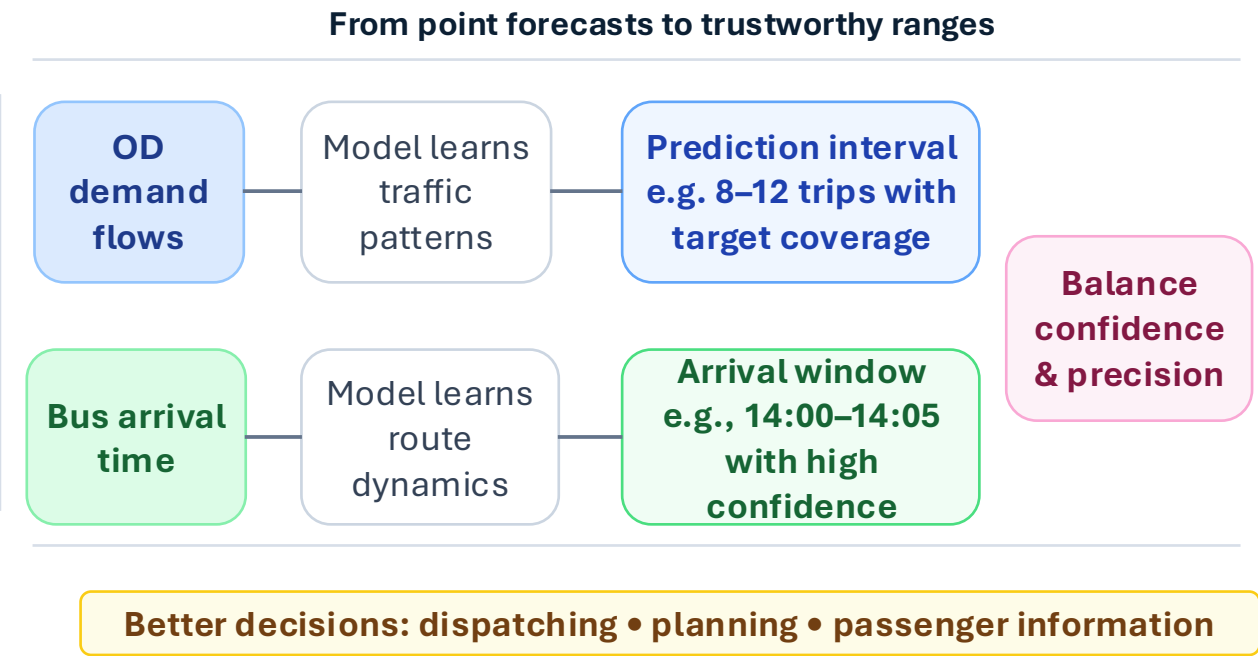
- Point forecasts hide risk: one prediction OD demand value / bus ETA can mislead dispatching, planning, and passenger decisions.
- Traffic data are variable: sparse OD flows and long bus prediction horizons need adaptive uncertainty windows.

3. Contribution / novelty

Two connected studies:

- OD demand: fair uncertainty-aware prediction that reports reliable possible demand ranges across magnitude-various demand groups.
- Bus arrival: arrival-time windows that are both reliable and precise for passengers and operators.

4. Method



5. Key takeaway: Uncertainty-aware traffic prediction turns one risky number into a decision-supported range, and checks whether the range is reliable across different situations.

Efficient Online OD Prediction via a Knowledge-driven Deep Learning Framework – Zeren XING

Project area: [Intelligent Transportation Systems/ OD Prediction] | Stage: [Journal Paper accepted]

1. Research aim

This research develops an efficient and explainable online origin-destination prediction framework that combines deep learning with gravity-model knowledge to predict real-time OD matrices.

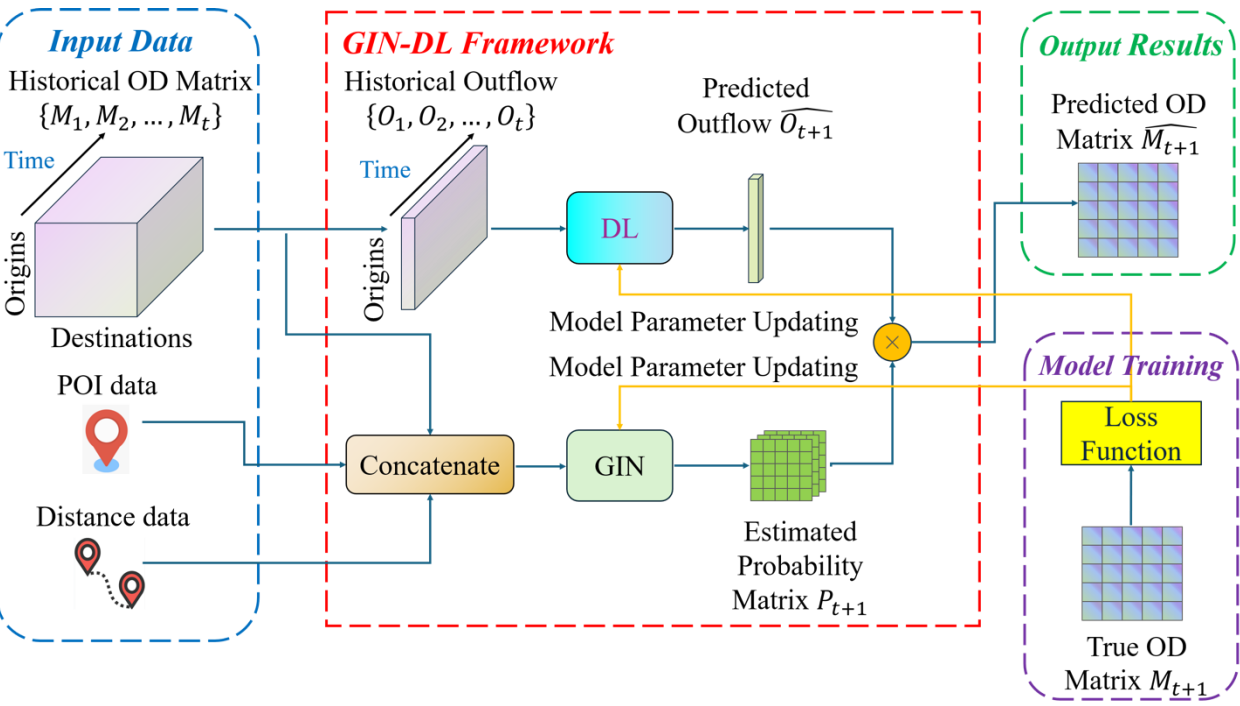
2. Why it matters

- Real-time OD prediction supports vehicle dispatch, rebalancing, and traffic management, but full OD matrices are costly to train because their size grows as N^2 .
- Existing deep learning-based OD prediction models are often slow to retrain, weakly adaptable to new data, and difficult for practitioners to understand.

3. Contribution / novelty

A knowledge-driven GIN-DL framework decomposes OD prediction into DL-based outflow prediction and gravity-inspired flow-probability estimation, plus a two-stage training strategy using keyframes and online fine-tuning.

4. Method



5. Key takeaway: The audience should remember that including transportation-domain knowledge into deep learning can make online OD prediction faster and more explainable while preserving competitive accuracy compared to deep learning-only models.

Simulation-based traffic signal coordination – Zhixian Tang

Project area: [Simulation/surrogate-based traffic signal optimization] | Stage: [submitted to journal]

1. Research aim

To develop a data-efficient, scalable, and interpretable surrogate model for simulation-based traffic signal coordination.

2. Why it matters

- Existing analytical models are overly simplified, without capturing some progression dynamics that progression performance is sensitive to.
- Existing end-to-end surrogates are black-box. They do not scale, interpret in traffic terms, and suffer from curse of dimensionality.

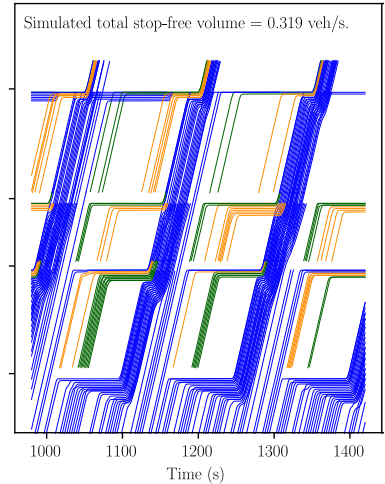
3. Contribution / novelty

- A structural surrogate model: using analytical progression model as backbone, only learning local, low-dimensional modules.
- A lightweight neural network (LWNN) as learner, which maintains MILP nature of the original analytical model.

4. Evidence visual

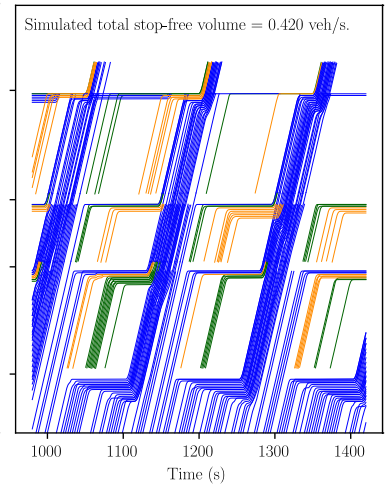
Intersection No.4 to No.1

Simulated total stop-free volume = 0.319 veh/s.



Intersection No.4 to No.1

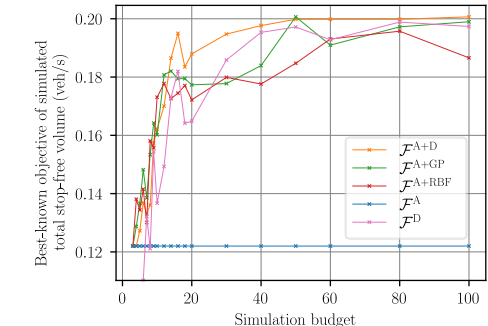
Simulated total stop-free volume = 0.420 veh/s.



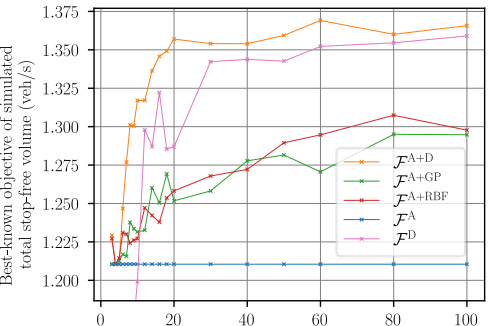
Analytical solution

Proposed solution

Vehicle trajectories



2-intersection network



7-intersection network

Scalability across network size

5. Key takeaway: Analytical models are used to organize surrogate representation, beyond their traditional role as a global objective estimation prior.

4

Multi-Leader Multi-Follower Stochastic Game for Large-Scale Traffic Signal Control – Di Chen

Project area: Large-Scale Traffic Signal Control / Stochastic Games / Multi-Agent Reinforcement Learning | Stage: Pre-submission revision

1. Research aim

This research develops a leader–follower framework for scalable traffic signal control, with critical intersections serving as leaders, and establishes equilibrium guarantees for the proposed multi-agent system.

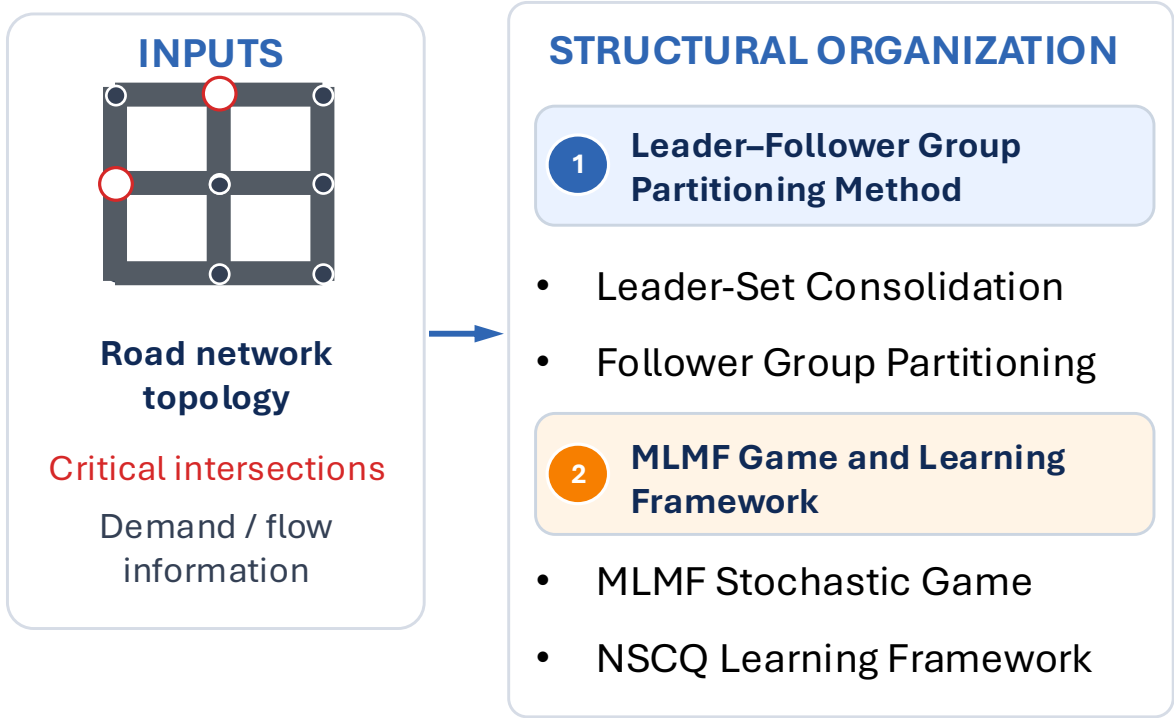
2. Why it matters

- Strongly coupled intersections require coordinated decisions, while full-network centralized control is difficult to scale.
- Learned signal policies need a formal way to explain whether their long-run interactions form a stable operating pattern.

3. Contribution / novelty

- Propose a partitioning algorithm to form leader–follower control groups.
- Develop an Multi-Leader Multi-Follower (MLMF) stochastic game and Nash-Stackelberg-Cooperative Q-learning(NSCQ) framework with equilibrium and convergence guarantees.

4. Framework



5. Key takeaway: The proposed leader–follower framework makes large-scale traffic signal control more scalable while using equilibrium to characterize policy stability.

Programmatic Policy Learning for Traffic Signal Control – Lindong Xie

Project area: Traffic signal control & Symbolic policy distillation | Stage: method + experiments + paper completed

1. Research aim

Learning programmatic policies from deep reinforcement learning (DRL) - based traffic signal controllers.

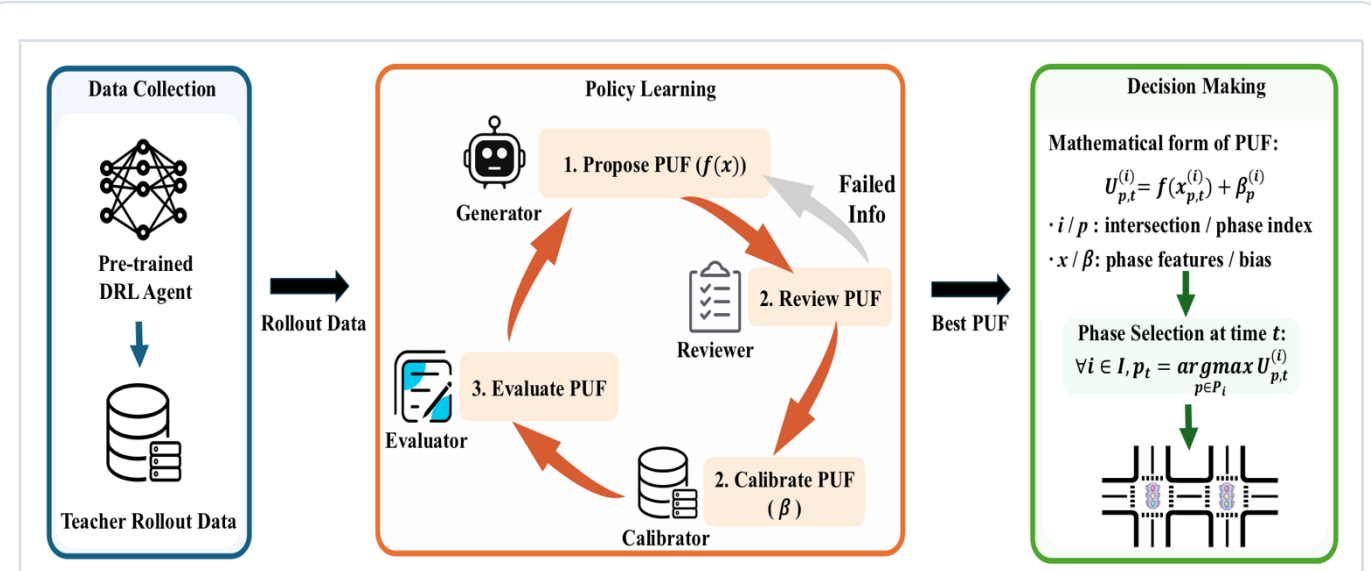
2. Why it matters

- DRL signal controllers are popular but opaque, computationally heavy, and hard to generalize.
- Programmatic policies are explicit and deployable, but difficult to learn from scratch.

3. Contribution / novelty

Develop a general programmatic policy learning framework that learns explicit signal control policies from DRL-based controllers.

4. Method / evidence visual



Evidence: +3.42% average in-city gain, +24.22% average cross-city transfer gain, and 1–3 orders lower storage than DRL controllers.

5. Key takeaway: Learned programmatic policies achieve competitive signal control performance, better transfer generalization, and much lower storage/computation cost than DRL controllers..

Safe Hybrid Multi-Agent Reinforcement Learning for On-Ramp Merging – Fan Tingting

1. Research aim

This work develops a safe hybrid multi-agent reinforcement learning and control barrier function (CBF) framework for hybrid decision-making in on-ramp merging for mixed CAV and HDV traffic.

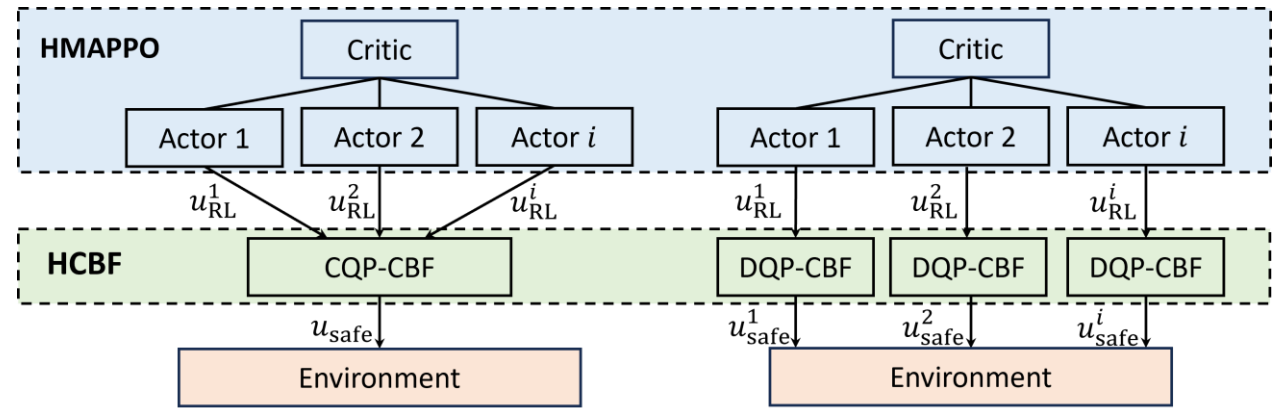
2. Research gap

- CBF method is limited to continuous dynamic systems.
- Safety-critical interactions in mixed CAV and HDV traffic remain underexplored.

3. Contribution

- Extend the forward invariance property of CBFs to hybrid dynamical systems, ensuring safety is preserved across both continuous dynamics & discrete lane-change transitions.
- Formulate hybrid multi-agent coordination as a Mixed-Integer Quadratic Program (MIQP) with pairwise hybrid CBF constraints to jointly optimize lane-change decisions & continuous control actions, with both centralized & decentralized manner.

4. Method / evidence visual



Centralized MIQP (CQP):

$$\min \sum_i^N \left((u_{safe}^i - u_{RL}^i)^2 + W \sum_{d \in [0,1,2]} \delta_{i,d} \right)_{\neq u_{RL,LC}^i}$$

Decentralized MIQP (DQP):

$$\min (u_{safe}^i - u_{RL}^i)^2 + W \sum_{d \in [0,1,2]} \delta_{i,d} \neq u_{RL,LC}^i$$

Mixed integer constraints for hybrid CBF:

$$\begin{aligned} \dot{h}^{SDH}(s, \mathbf{u}_i, \hat{\mathbf{u}}_j) + \alpha h^{SDH}(s) &\geq -M * (1 - \delta_{i,d}) && \text{CAV-Leader pair} \\ \dot{h}^{TTC}(s, \mathbf{u}_i, \hat{\mathbf{u}}_j) + \alpha h^{TTC}(s) &\geq -M * (1 - \delta_{i,d}) && \text{CAV-Follower pair} \\ \dot{h}^{Ramp}(s, \mathbf{u}_i) + \alpha h^{Ramp}(s) &\geq -M * (1 - \delta_{i,0}) && \text{Ramp lane end} \\ a_{min} \leq u_{safe}^i &\leq a_{max} && \text{Control bound} \end{aligned}$$

RL & LLM for Large-scale EVs Charging Scheduling – Yang Zhang

Project area: [Reinforcement Learning & Large Language Model] Stage: [Submitted to IEEE Internet of Things (Reject & Resubmit)]

1. Research aim

Integrate RL & LLM for large-scale EVs charging scheduling involving three stakeholders perspectives: EVs, charging stations, & the power grid, to balance the three user types' benefits & improve charging market operation efficiency.

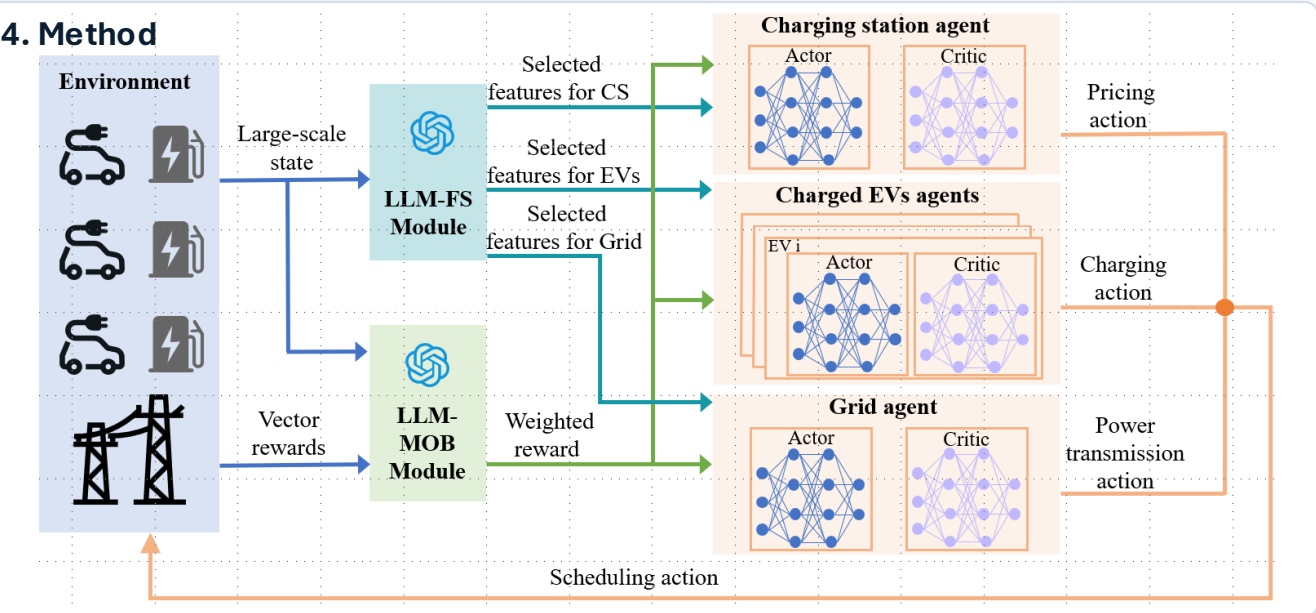
2.Problems to address

- The vast state feature space arising from complex multi-stakeholder interactions, creating computational bottlenecks & hinders effective policy learning.
- Conflicting objectives among stakeholders, where optimizing one party's benefits often deteriorates others' performance, making unified optimization difficult.

3. Contribution

- RL-driven unified EV-Station-Grid operation model.
- LLM-based interpretable feature selection to assess feature importance based on their semantic relevance, and contribution to stakeholder objectives .
- LLM-based adaptive multi-objective balancing to dynamically assign weights to conflicting objectives without requiring complex Pareto frontier maintenance.
- Integrate the above LLM modules with MADDPG, enabling the efficient learning of high-quality, balanced scheduling policies for the unified market.

4. Method



5. Results

LLM Feature Selection Output: EV Perspective

1. **Desired SoC for charged EVs** (1.0)
Critical as it directly influences the completion of the charging session and satisfaction of charging needs
2. **Desired departure time for charged EVs** (0.95)
Very important for scheduling and minimizing waiting times and charging times
3. **Charging and waiting time cost weight factors** (0.9)
These influence user satisfaction and optimization of time-based objectives
4. **Energy stored in the battery system of CS** (0.8)
Determines available resources for charging and affects time reduction
5. **Charging efficiency for charged EVs** (0.75)
Directly impacts the charging time and energy required

LLM Multi-Objective Optimization Reasoning

Reasoning: With 18 EVs currently being charged, several have imminent departure times (one at 7 o'clock and several at 8 o'clock), indicating pressing needs to optimize charging power and minimize charging/waiting time. Since EV drivers are sensitive to charging times and costs, these should be prioritized. Given the variety of desired SOC levels and departure times, the focus should be on ensuring that EVs meet their energy requirements efficiently and affordably. The diverse coefficients signify urgency and variability in how charging time costs impact different vehicles, suggesting dynamic control of charging power to accommodate EV needs.

Decision: weight 1.5 for EVs standpoint.

EV Charging Pricing for Congestion Management – Gaojunjie Li

Project area: Transportation electrification, electricity markets, coupled traffic-power systems | Stage: validation / writing

1. Research aim

To develop a **real-time coordinated pricing framework** for multiple electric vehicle (EV) charging stations in coupled traffic-power networks **to mitigate traffic congestion** in high-density cities.

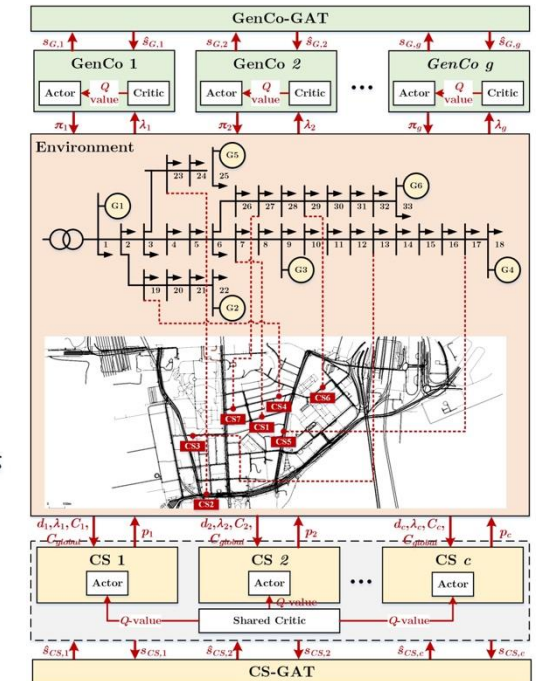
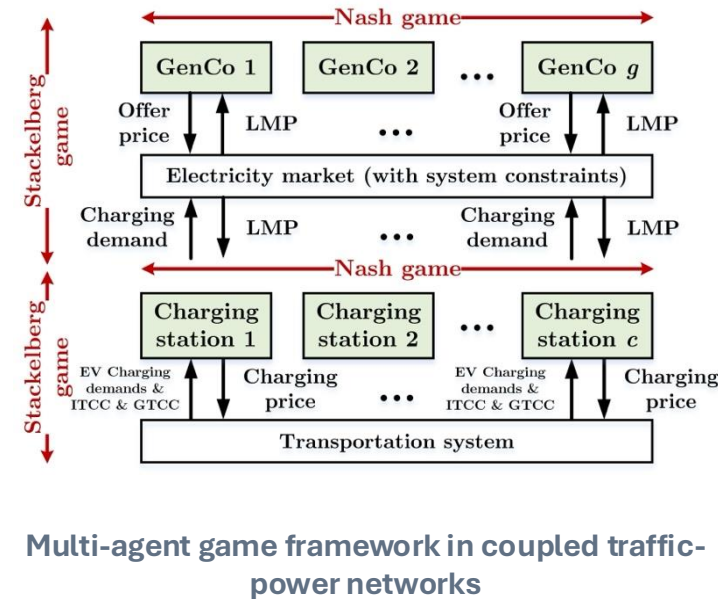
2. Why it matters

- Existing charging station (CS) pricing rarely internalizes **real-time traffic congestion costs**.
- Coupled generation company (GenCo)-CS-EV interactions make pricing a **multi-agent multi-level problem**.

3. Contribution / novelty

- A **SUMO-based charging pricing framework** is established to embed fine-grained congestion-cost feedback from real-time traffic dynamics into CS decisions.
- A **graph attention network-based bi-level multi-agent reinforcement learning algorithm** is proposed to coordinate GenCo bidding and CS pricing.

4. Method



5. Key takeaway: Charging prices can serve as real-time congestion management signals, shaping EV charging demand and traffic flows while coordinating with electricity market bidding in coupled traffic-power networks.

Planning and Operation of Future Railway System with Autonomous Modularized Trains and Virtual Coupling Signaling – Maoning CHEN

Project area: [Railway, Scheduling, Optimization] | Stage: [writing]

1. Research aim

This research aims at **scheduling** autonomous modularized trains under **virtual coupling signaling**, considering multiple operational decisions.

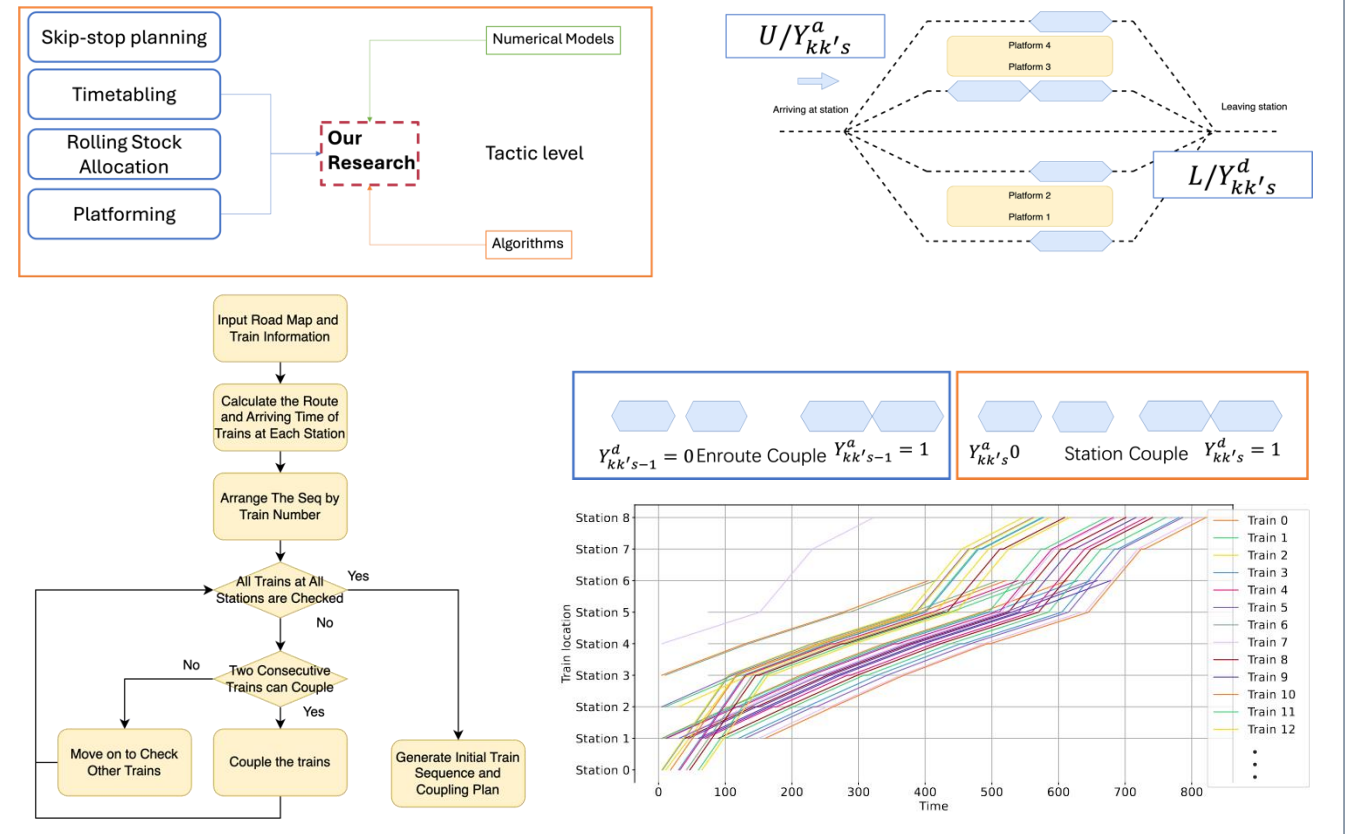
2. Why it matters

- Virtual coupling signaling is a new control method that can enlarge the capacity of trains.
- A general and comprehensive **model** with effective **algorithm** is required to make precise decisions.

3. Contribution / novelty

- Scheduling Virtual Coupling Train System under multiple planning elements including overtaking, platforming, coupling and decoupling.
- Considering convergence and separation of trains.
- Using Mixed Integer Programming approach.
- Heuristics algorithms and exact algorithms.

4. Method



5. Key takeaway:

“In a virtual coupling train system, the highly interrelated decisions are made in stages, with the solution space reduced through various valid inequalities and cuts.”

Integrated Airport Surface Planning via Logic-Based Benders Decomposition – HAN

Yihang

Project area: [Airport operations optimization / integrated airport surface planning] | Stage: [Algorithm design]

1. Research aim

This research develops an integrated pre-planning optimization framework for airport gate assignment, taxiway planning, and runway scheduling, with a customized logic-based Benders decomposition algorithm for efficient solution.

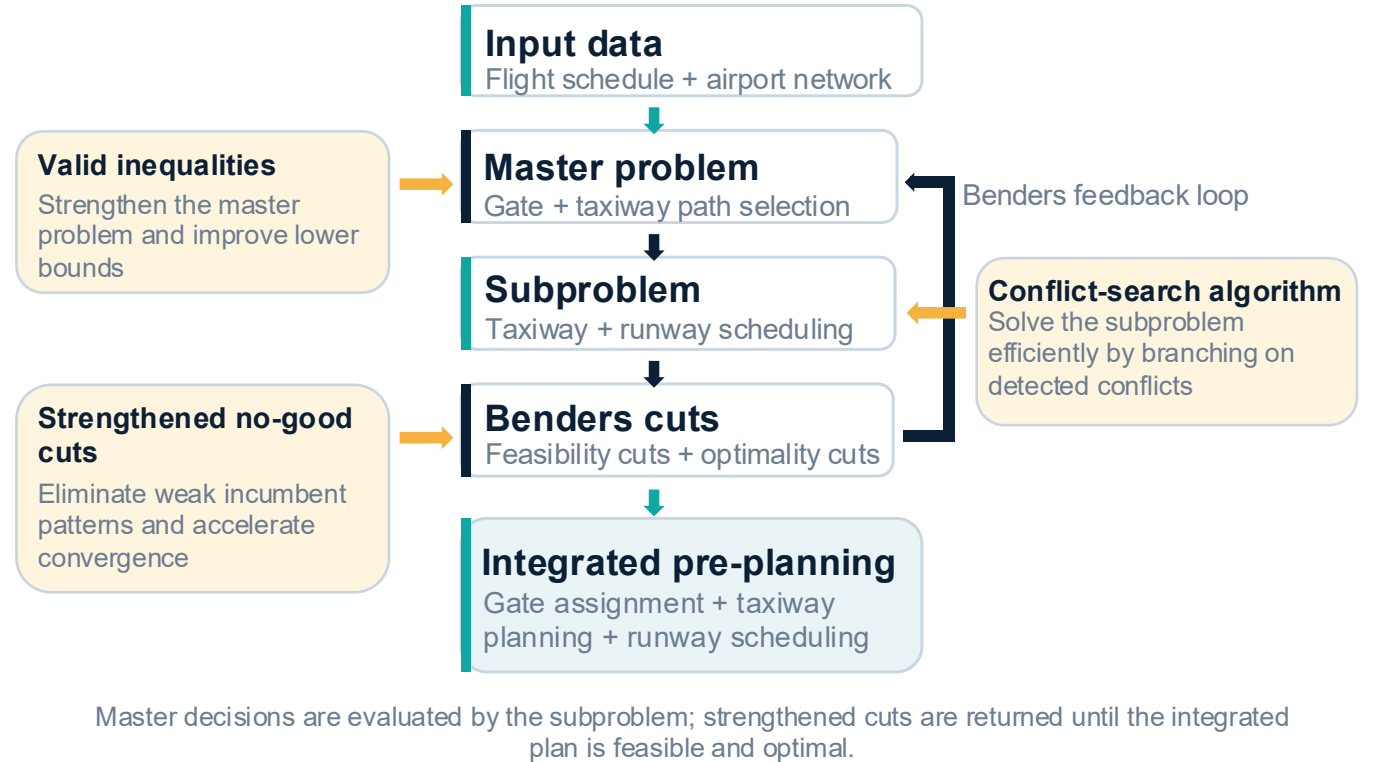
2. Why it matters

- Separate planning may ignore important interactions among gates, taxiways, and runways.
- Integrated planning improves coordination but creates a large-scale and computationally difficult optimization problem.

3. Contribution / novelty

A **logic-based Benders decomposition** framework is designed for the integrated problem. Several **acceleration strategies** are proposed, including valid inequalities, a conflict-search-based subproblem solution algorithm, and strengthened no-good cuts.

4. Method



5. Key takeaway: A tailored logic-based Benders decomposition algorithm enables efficient integrated airport surface planning.